

Physical Chemistry V:

Spectroscopy of atoms and molecules

Spectroscopy is the study of absorption and emission of radiation by matter.

Goal: Determine the structural and dynamical properties of matter.

Content

- 1 Introduction
- 2 Atomic structure and spectra
- 3 Structure and spectra of diatomic molecules
- 4 Group theory
- 5 Structure and spectra of polyatomic molecules

Prerequisites

ACPC I: Introduction to quantum mechanics

PC III: Molecular quantum mechanics

Repetition of important concepts from ACPC I and PC III:

<https://moodle-app2.let.ethz.ch/course/view.php?id=17298>

Chapter 1

Introduction

1.1 Underlying concepts

The Nobel Prize in Chemistry 1971

Gerhard Herzberg

Gerhard Herzberg

Born: 25 December 1904, Hamburg, Germany

Died: 3 March 1999, Ottawa, Canada

Affiliation at the time of the award:

National Research Council of Canada, Ottawa, Canada

Prize motivation: "for his contributions to the knowledge of electronic structure and geometry of molecules, particularly free radicals"

Field: Physical chemistry, molecular structures



“...series of bands with complicated fine structure in the red and photographic infrared...
This spectrum was analysed as that of an asymmetric top...” (1)

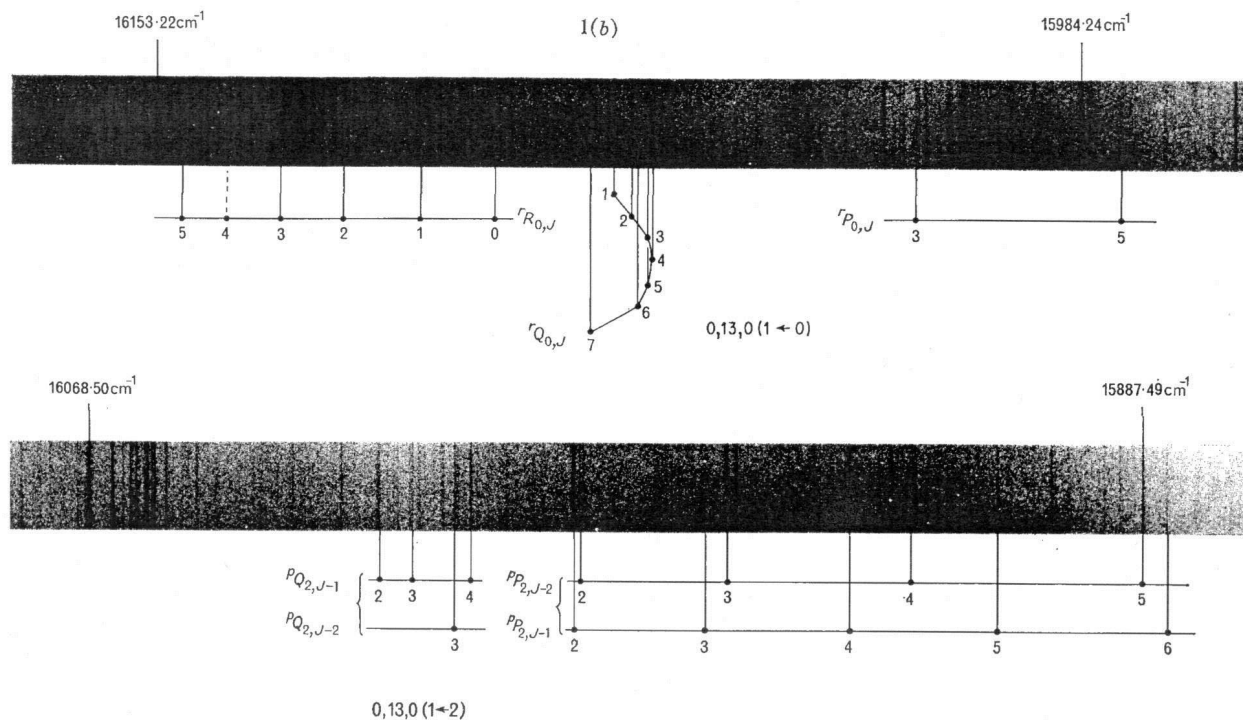
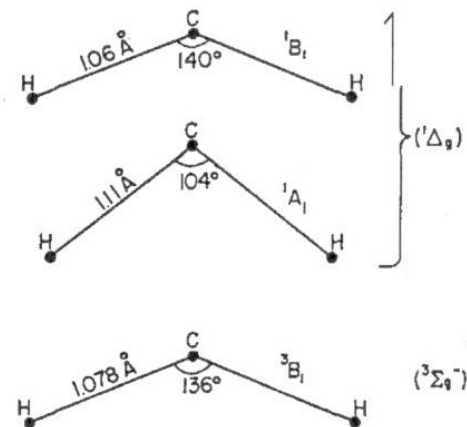
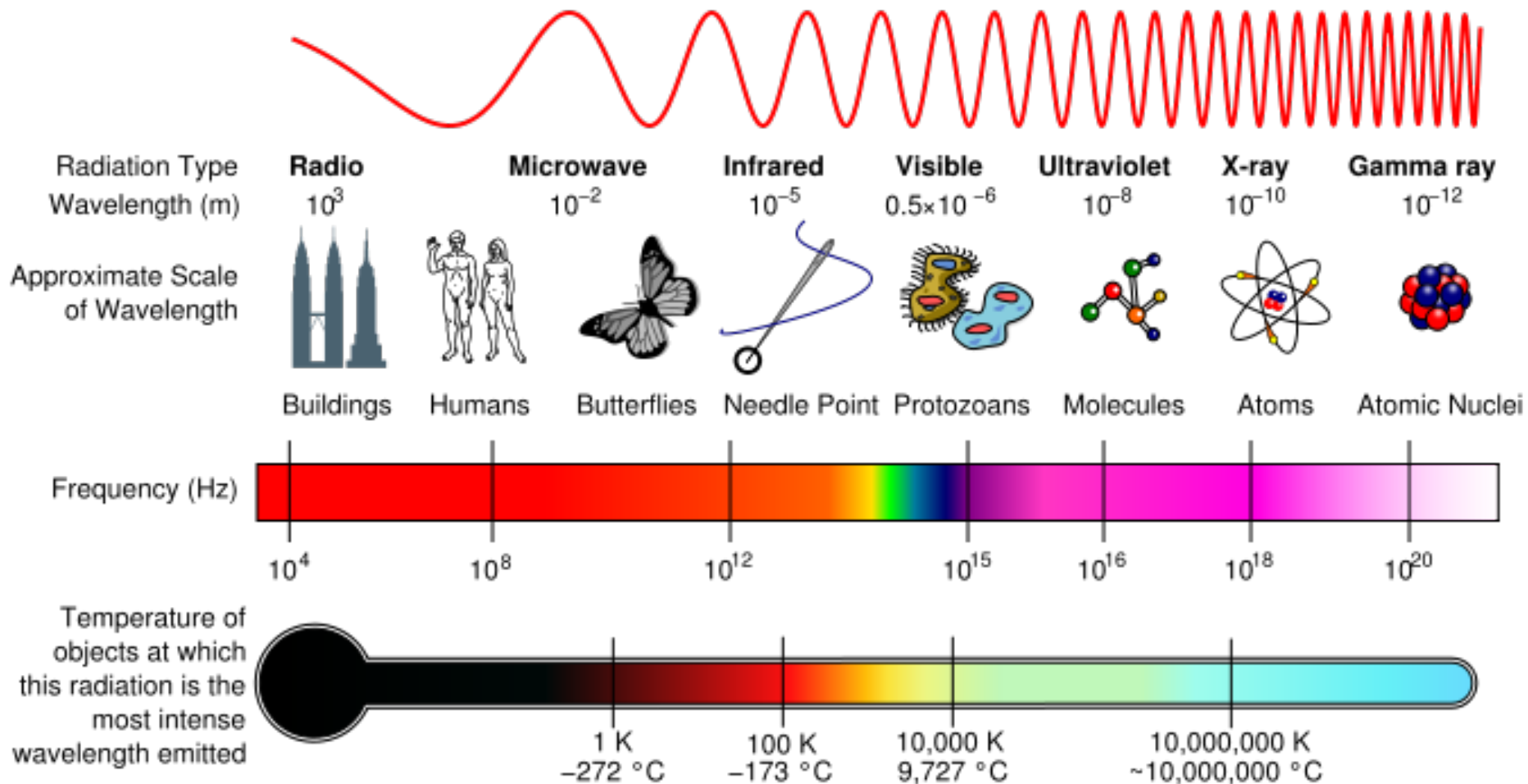


Fig. 18.
Two subbands (1-0 and 1-2) of the 0 13 O-O 0 0 band of the red absorption system
of CH_2 .

Geometrical structure of CH_2
in the three lowest states

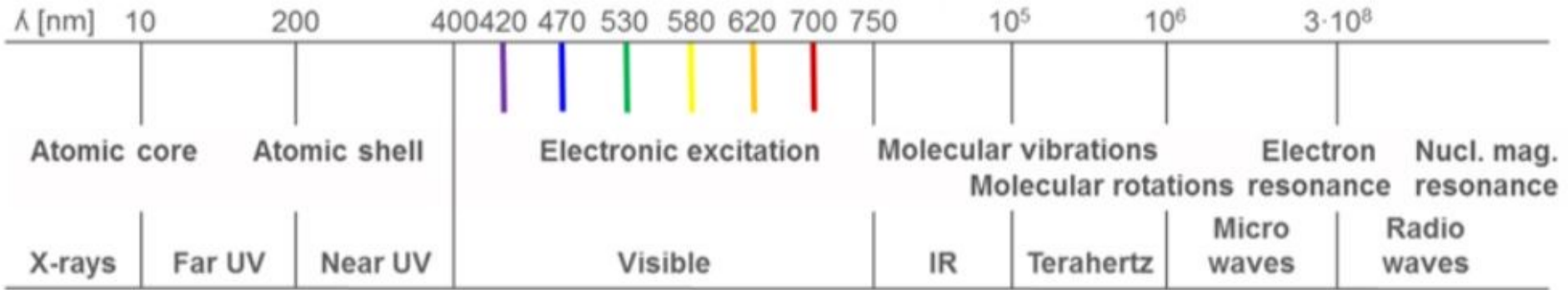


The electromagnetic spectrum

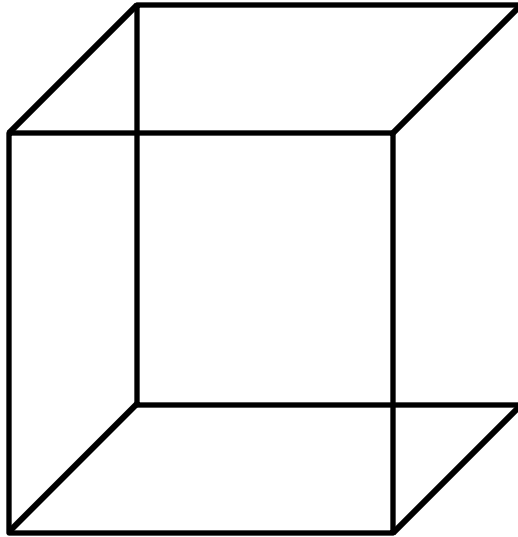


Adapted from en.wikipedia.org/wiki/Electromagnetic_spectrum

Spectral regions and types of spectroscopy



1.2 Thermal radiation and Planck's law



L

Cubic cavity

Thermal equilibrium:

Emitted power = absorbed power at all ν

$$P_e(\nu) = P_a(\nu)$$

General form of the electric component of electromagnetic waves:

$$E(\vec{r}, t) = \sum_p A_p \exp \left[i(\underbrace{\omega_p t}_{\text{angular frequency}} - \underbrace{\vec{k}_p \cdot \vec{r}}_{\text{wave vector}}) \right] + \text{compl. conj.}$$

angular frequency

wave vector

The electric field strength has to be zero on the walls of the cavity which imposes the boundary condition

$$E(x = 0, y, z, t) = E(x = L, y, z, t) = 0 \quad \text{with } 0 \leq y \leq L \quad \text{and} \quad 0 \leq z \leq L \quad (1.2)$$

and similar conditions for the y and z dimensions. Consequently, the electric field takes the form

$$E(\vec{r}, t) = A \sin(k_x x) \sin(k_y y) \sin(k_z z), \quad (1.3)$$

which imposes the boundary conditions on the wave vector

$$k_x = \frac{\pi}{L}n_1, k_y = \frac{\pi}{L}n_2, k_z = \frac{\pi}{L}n_3, \quad (1.4)$$

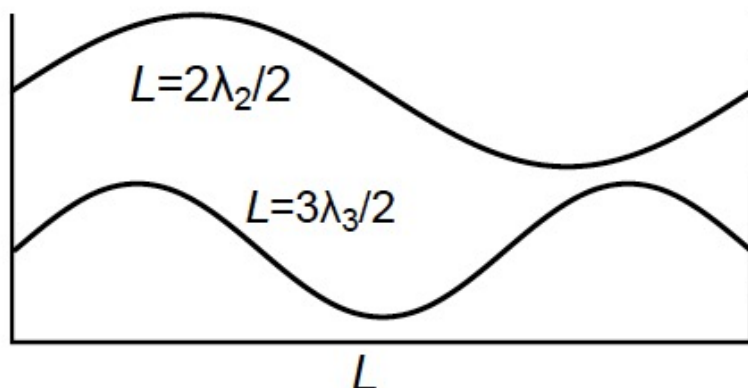


Figure 1.2: Allowed electro-magnetic waves in a cavity of length L .

The wave vector of the radiation field can thus be written as

$$\vec{k} = \frac{\pi}{L}(n_1, n_2, n_3)$$

with positive integers n_i ($n_i \in \mathbb{N}^*$). The magnitude of the allowed wave vectors is thus

$$|\vec{k}| = \frac{\pi}{L}\sqrt{n_1^2 + n_2^2 + n_3^2}$$

which can be written in terms of the wavelength $\lambda = 2\pi/|\vec{k}|$, the frequency $\nu = c|\vec{k}|/2\pi$ or the angular frequency $\omega = c|\vec{k}|$.

The modes with angular frequency between 0 and ω fulfils

$$n_1^2 + n_2^2 + n_3^2 \leq \frac{\omega^2 L^2}{c^2 \pi^2}. \quad (1.7)$$

Equation (1.7) represents the interior of a sphere with the radius

$$r = \sqrt{n_1^2 + n_2^2 + n_3^2} = \frac{\omega L}{c\pi}. \quad (1.8)$$

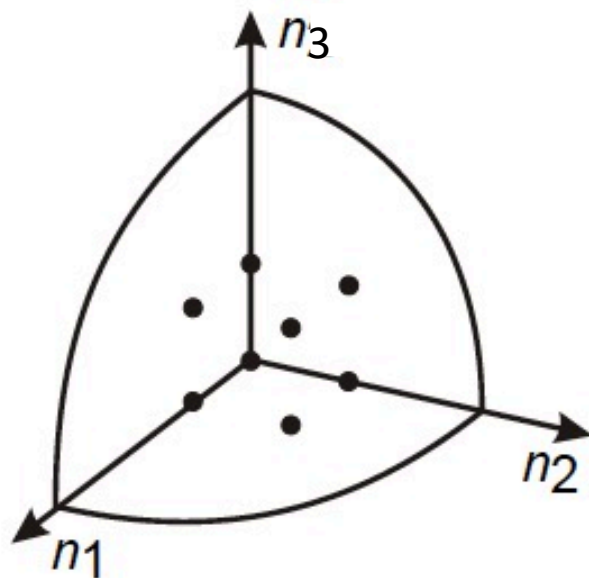


Figure 1.3: Diagram of the possible (n_1, n_2, n_3) combinations for occupation numbers of the modes of an optical cavity. The possible combinations are contained within an octant of a sphere of radius $r = \frac{\omega L}{c\pi}$.

In this limit, the number of modes can be estimated from the volume of the octant of the sphere shown in Figure 1.3. This volume is

$$V_{\text{oct}} = \frac{1}{8} \frac{4\pi}{3} \left(\frac{\omega L}{c\pi} \right)^3. \tag{1.9}$$

Since two polarization states are possible for each mode, this leads to the mode number

$$N_{\omega}(\omega) = 2V_{\text{oct}} = \frac{\pi}{3} \left(\frac{\omega L}{c\pi} \right)^3 \tag{1.10}$$

and, expressed in frequencies $\nu = \frac{\omega}{2\pi}$,

$$N(\nu) = \frac{8\pi\nu^3 L^3}{3c^3}. \tag{1.11}$$

Density of modes per unit volume

$$g(\nu) = \frac{1}{V} \frac{dN(\nu)}{d\nu} = \frac{8\pi\nu^2}{c^3}.$$

Each electromagnetic mode of the cavity is in thermal equilibrium with the matter of the cavity walls.

Matter can be described as a series of classical mechanical oscillators at temperature T
Mean energy: kT

Mean energy in each mode is also kT .

Energy content of radiation field is:

$$U(\nu)d\nu = g(\nu)VkTd\nu = \frac{8\pi\nu^2}{c^3}VkTd\nu.$$

Energy density is:

$$\rho(\nu) = \frac{U(\nu)}{V} = g(\nu)kT = \frac{8\pi\nu^2}{c^3}kT.$$

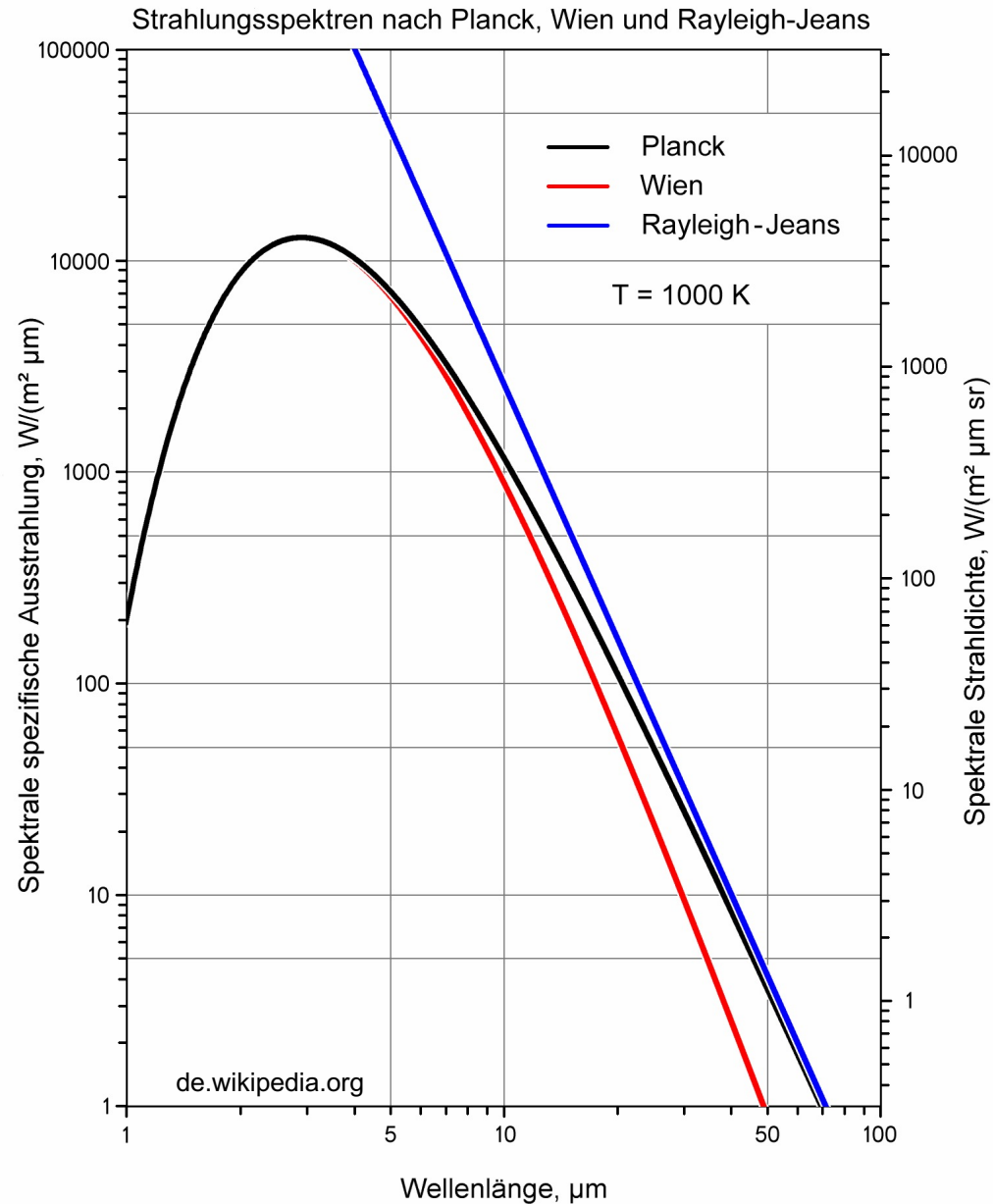
Known as “Rayleigh-Jeans law”.

It is only correct in the limit of low frequencies $h\nu \ll kT$ and long wavelengths (infrared).

In the limit of short wavelengths (ultraviolet), the energy density diverges!

This was called the “UV catastrophe”.

Radiation Laws



Max Planck

Max Karl Ernst Ludwig Planck

Born: 23 April 1858, Kiel, Schleswig

Died: 4 October 1947, Göttingen, West Germany

Affiliation at the time of the award:

Berlin University, Berlin, Germany

Prize motivation: "in recognition of the services he rendered to the advancement of Physics by his discovery of energy quanta"

Field: Quantum mechanics



Planck's law

- Hypothesis:
Each mode of the radiation field can only emit or absorb energy in integer multiples q of a fundamental energy quantum $h\nu$, where h is Planck's constant.
- This quantum is nowadays called “photon”.
- Energy carried by a photon can be expressed as follows:

$$E = h\nu = \hbar\omega = \frac{hc}{\lambda} = hc\tilde{\nu}$$

- Approximate conversion factors that are useful to remember:

$$1.000 \text{ eV} \leftrightarrow 2.418 \times 10^{14} \text{ Hz} \leftrightarrow 8066 \text{ cm}^{-1}$$

- Partitioning of energy in thermal equilibrium is given by Maxwell-Boltzmann distribution:
- Probability $p(q)$ to find the energy $qh\nu$ in a specific mode is:

$$p(q) = \frac{1}{Z} \exp\left(-\frac{qh\nu}{kT}\right)$$

where $Z = \sum_q \exp\left(-\frac{qh\nu}{kT}\right)$ is the “partition function”, ensuring $\sum_q p(q) = 1$

Planck's law (continued)

- Mean energy per mode is:

$$\langle E(\nu) \rangle = \sum_{q=0}^{\infty} p(q) q h \nu = \frac{\sum_q q h \nu \exp(-q h \nu / (kT))}{\sum_q \exp(-q h \nu / (kT))}$$

leading to:

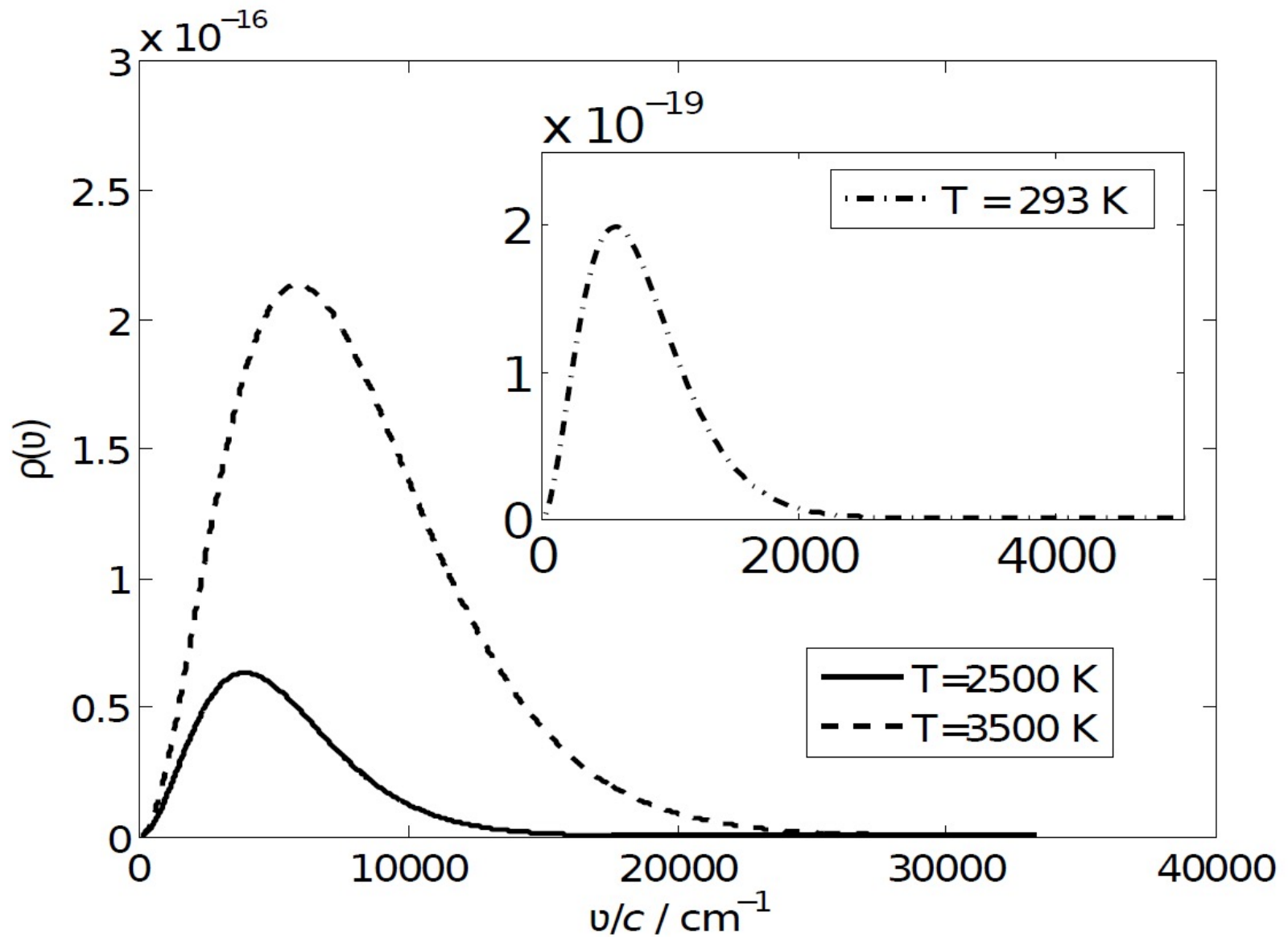
$$\langle E(\nu) \rangle = \frac{h \nu}{\exp(h \nu / (kT)) - 1}$$

- Energy density of radiation field within the interval ν to $\nu + d\nu$ is given by:

$$\rho(\nu) d\nu = g(\nu) \langle E(\nu) \rangle d\nu = \frac{8\pi \nu^2}{c^3} \frac{h \nu}{\exp(h \nu / (kT)) - 1} d\nu.$$

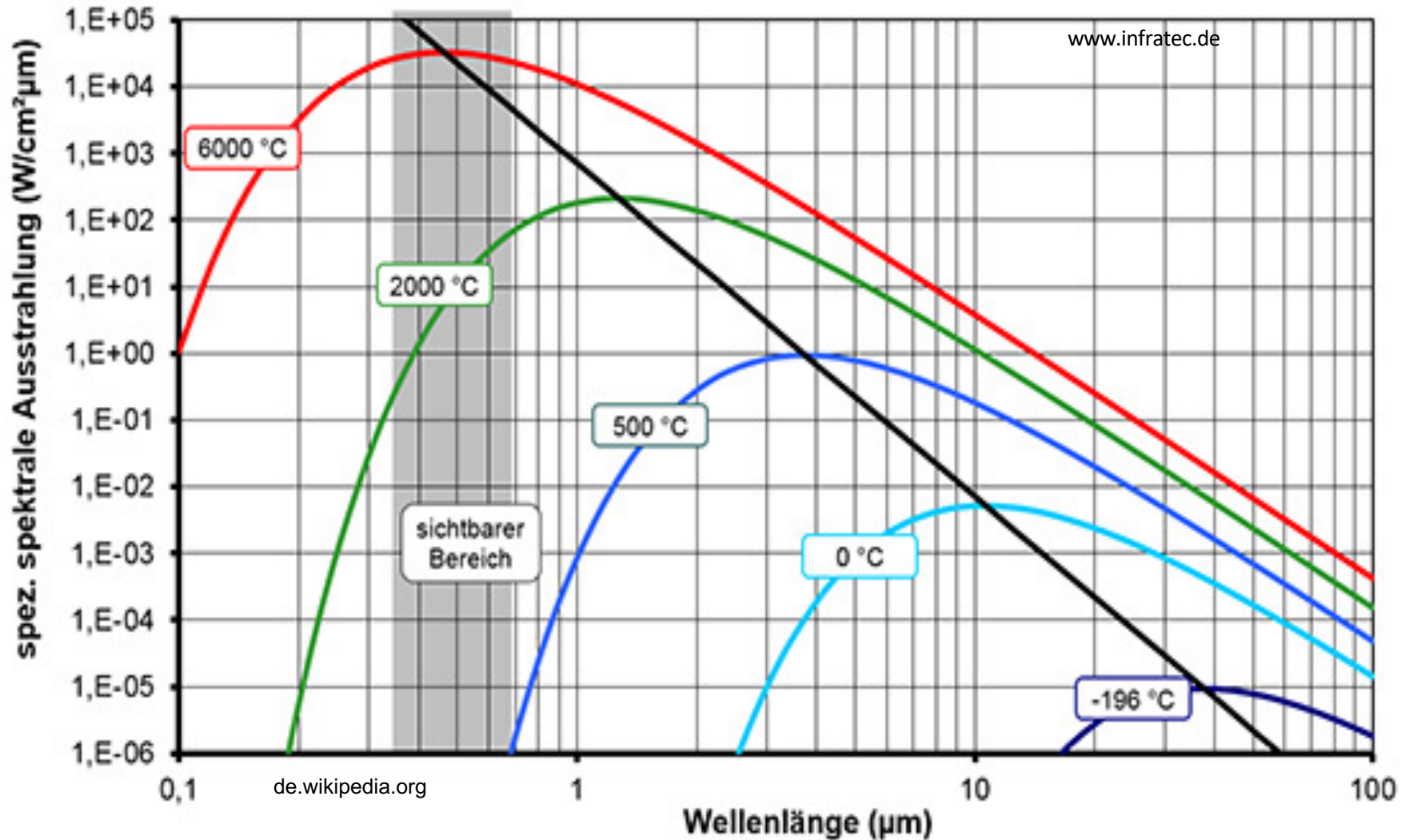
This is known as “Planck’s law”.

Planck's law



Planck's law

Strahlungsspektren eines «schwarzen Körpers»

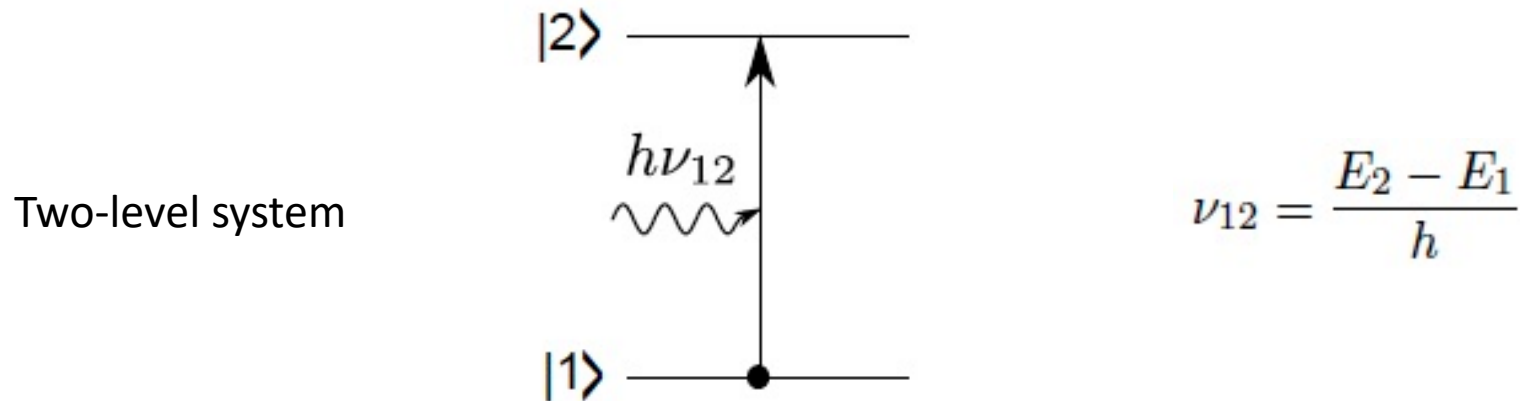


Planck's law

T/K	$\nu_{\text{max}}/\text{Hz}$	$\tilde{\nu}_{\text{max}}/\text{cm}^{-1}$	$\lambda(\nu_{\text{max}})/\mu\text{m}$	region
3	1.76×10^{11}	5.88	1700	cosmic background
300	1.76×10^{13}	588	17	surface of the earth
6000	3.53×10^{14}	11768	0.85	surface of the sun

Table 1.1: Frequencies ν_{max} , wavenumbers $\tilde{\nu}_{\text{max}}$ and wavelengths $\lambda(\nu_{\text{max}})$ describing the maximum of the energy density distribution $\rho(\nu)$ for different temperatures T .

1.3 Absorption and emission of radiation



Kinetic formalism: The rates of change of the populations N_i in the two states are:

$$-\frac{dN_1^{(\text{abs})}}{dt} = \frac{dN_2^{(\text{abs})}}{dt} = B_{12}\rho(\nu_{12})N_1$$

Depends on properties of the (two-level) system. Photon density

Albert Einstein

Albert Einstein

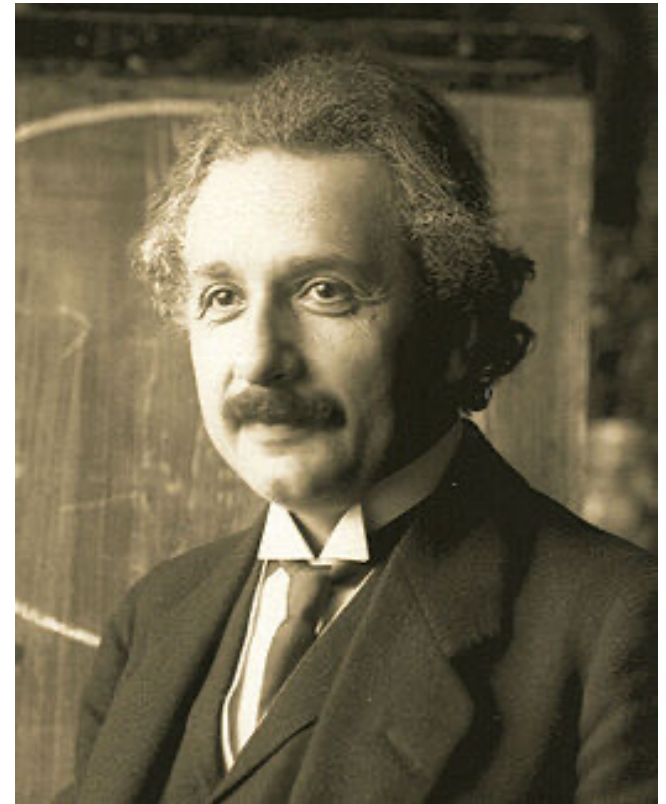
Born: 14 March 1879, Ulm, Germany

Died: 18 April 1955, Princeton, NJ, USA

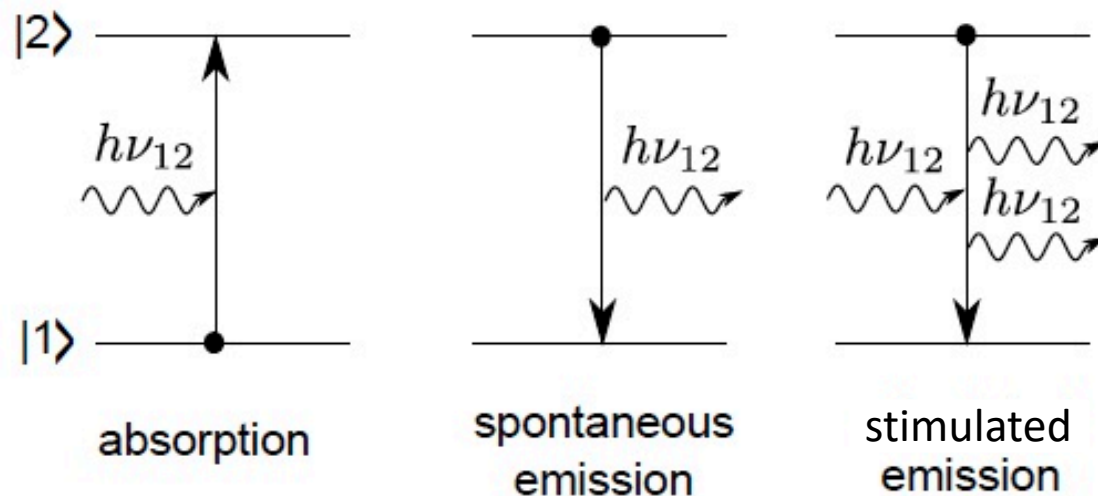
Affiliation at the time of the award: Kaiser-Wilhelm-Institut (now Max-Planck-Institut) für Physik, Berlin, Germany

Prize motivation: "for his services to Theoretical Physics, and especially for his discovery of the law of the photoelectric effect"

Field: Theoretical physics



There are 3 elementary processes:



Spontaneous emission:

$$\frac{dN_1^{(\text{sp. em.})}}{dt} = -\frac{dN_2^{(\text{sp. em.})}}{dt} = A_{21}N_2$$

Stimulated emission:

$$\frac{dN_1^{(\text{stim. em.})}}{dt} = -\frac{dN_2^{(\text{stim. em.})}}{dt} = B_{21}\rho(\nu_{12})N_2$$

- Two-level system is assumed to be in equilibrium with a thermal radiation field
 → Rate of absorption and total emission must balance each other

$$\left(\frac{dN_1}{dt} \right)_{\text{eq}} = -B_{12}\rho(\nu_{12})N_1^{(\text{eq})} + A_{21}N_2^{(\text{eq})} + B_{21}\rho(\nu_{12})N_2^{(\text{eq})} = 0$$

or

$$B_{12}\rho(\nu_{12})N_1^{(\text{eq})} = [A_{21} + B_{21}\rho(\nu_{12})] N_2^{(\text{eq})}$$

- Thermal equilibrium with environment at temperature T additionally requires:

$$\frac{N_2^{(\text{eq})}}{N_1^{(\text{eq})}} = \exp \left(-\frac{E_2 - E_1}{kT} \right) = \exp \left(-\frac{h\nu_{12}}{kT} \right)$$

giving

$$\rho(\nu_{12}) = \frac{A_{21}/B_{21}}{\left(\frac{B_{12}}{B_{21}} \right) \exp \left(\frac{h\nu_{12}}{kT} \right) - 1}$$

- Two equations must be simultaneously fulfilled at the same T :

$$\rho(\nu) = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{\exp(h\nu/(kT)) - 1} \quad \text{and} \quad \rho(\nu_{12}) = \frac{A_{21}/B_{21}}{\left(\frac{B_{12}}{B_{21}}\right) \exp\left(\frac{h\nu_{12}}{kT}\right) - 1}$$

leading to:

$$B_{12} = B_{21} \quad \text{and} \quad A_{21} = \frac{8\pi h\nu_{12}^3}{c^3} B_{21}$$

Implications:

- Einstein coefficients for absorption and stimulated emission are equal!
- The rate of spontaneous emission scales with ν^3 .

Einstein coefficients can be related to the wave functions of the two states. Within the dipole approximation, we will show (later) that

$$B_{12} = \frac{8\pi^3}{3h^2(4\pi\epsilon_0)} \left| \langle 2 | \hat{\mu} | 1 \rangle \right|^2$$

Average number of photons in each mode at temperature T :

$$n(\nu, T) = \frac{\langle E(\nu) \rangle}{h\nu} = \frac{1}{\exp(h\nu/(kT)) - 1}$$

At room temperature, in the visible range ($\lambda=600\text{ nm}$), one finds $n(\nu_{\text{vis}}, T_{\text{amb}}) \approx 2.5 \times 10^{-36}$

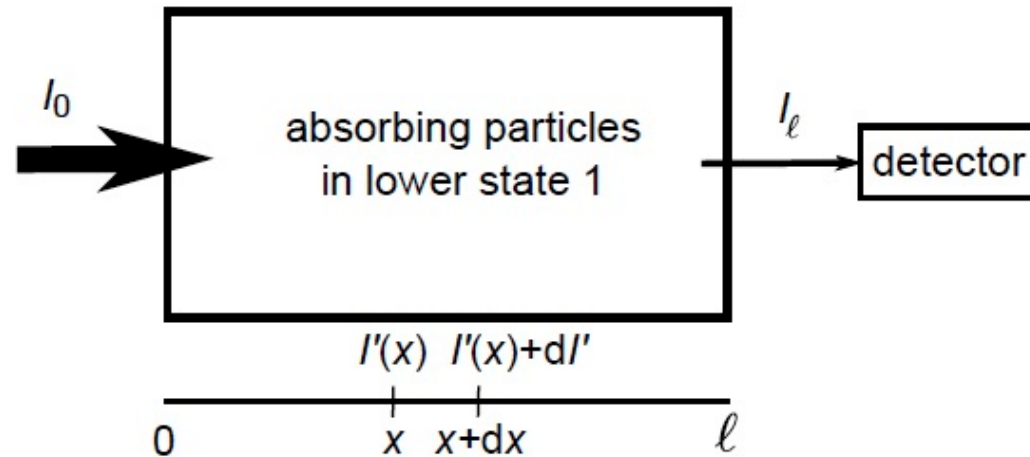
Therefore, spontaneous emission exceeds by far the thermally stimulated emission.

Exceptions:

- In a laser, it is possible to concentrate the radiation into a small number of modes, leading to occupation numbers $n(\nu, T)$ much larger than one.
- In the limit of high T , $\exp(h\nu/(kT)) \rightarrow 1$, which can also lead to very large occupation numbers.

1.4 Measurable quantities in spectroscopy

Goal: Establish a link between measurable quantities and Einstein coefficients.



Flow of photons per unit surface: $I = cC_\gamma$ C_γ is the photon concentration

Differential decrease of intensity: $-dI'(x) = \sigma_{12}I'(x)Cdx$

σ_{12} has the dimension of a surface and is called the absorption cross section

$$\text{L} \rightarrow - \int_{I_0}^{I_\ell} \frac{dI'}{I'} = \int_0^\ell \sigma_{12} C dx \quad \rightarrow \quad \ln \left(\frac{I_0}{I_\ell} \right) = \sigma_{12} C \ell$$

“Lambert-Beer law”

Description of absorption as a bimolecular reaction between particles of concentration C and photons of concentration C_γ .

$$-\frac{dC_\gamma}{dt} = k_{\text{bi}} C_\gamma C$$

Transforming position to time using the speed of light: $c = \frac{dx}{dt}$

$$-\frac{dI(x)}{dx} = -\frac{dC_\gamma c}{dx} = -\frac{dC_\gamma}{dx} \frac{dx}{dt} = -\frac{dC_\gamma}{dt}$$

The absorbing species can be considered to be at rest relative to the photons:

$$-\frac{dC_\gamma}{dt} = k_{\text{bi}} C_\gamma C = \sigma_{12} c C_\gamma C$$

➡ Bimolecular rate constant: $k_{\text{bi}} = \sigma_{12} c$

Pseudo-first-order rate constant (assuming C_γ constant): $k_{\text{first}} = \sigma_{12} c C_\gamma$

Comparison to rate coefficients obtained from Einstein coefficients:

$$K_{12} = B_{12} \rho(\nu_{12}) \quad K_{21} = B_{21} \rho(\nu_{12})$$

Quantitative measurements of absorption cross sections

All measured lines in spectroscopy have a finite linewidth caused by:

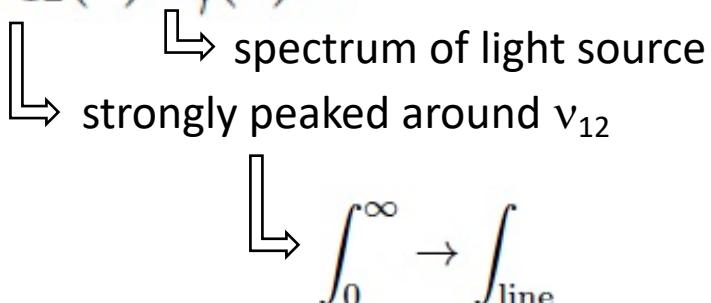
- Doppler broadening
- power broadening
- finite resolution of instruments
- ...

We therefore consider photon concentration in frequency interval ν to $\nu+d\nu$:

$$dC'_\gamma = C'_\gamma(\nu)d\nu \quad \text{with} \quad C_\gamma = \int_0^\infty C'_\gamma(\nu)d\nu.$$

Rate constants are obtained by integrating over all frequencies of transition $2 \leftrightarrow 1$

$$K_{12} = \int_0^\infty K_{12}(\nu)d\nu = \int_0^\infty c\sigma_{12}(\nu)C'_\gamma(\nu)d\nu$$



strongly peaked around ν_{12}

$\int_0^\infty \rightarrow \int_{\text{line}}$

$C'_\gamma(\nu)$ is related to the energy density $\rho(\nu)$ through $C'_\gamma(\nu) = \frac{\rho(\nu)}{h\nu}$ and therefore,

$$K_{12} = \frac{c}{h} \int_{\text{line}} \frac{\sigma_{12}(\nu)}{\nu} \rho(\nu) d\nu.$$

When the photon density of the excitation source varies little with its frequency

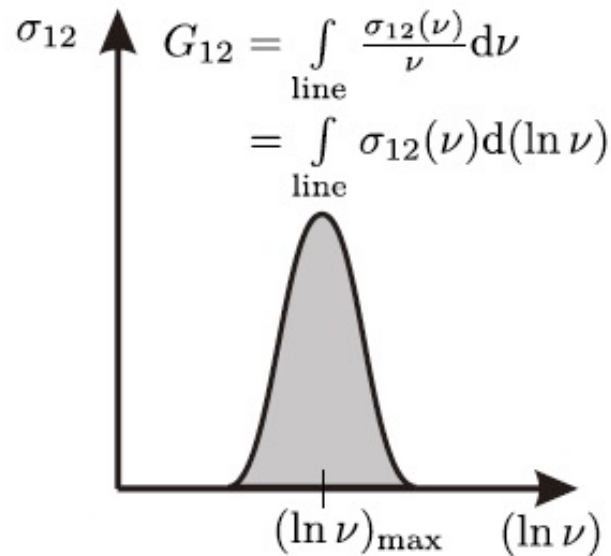
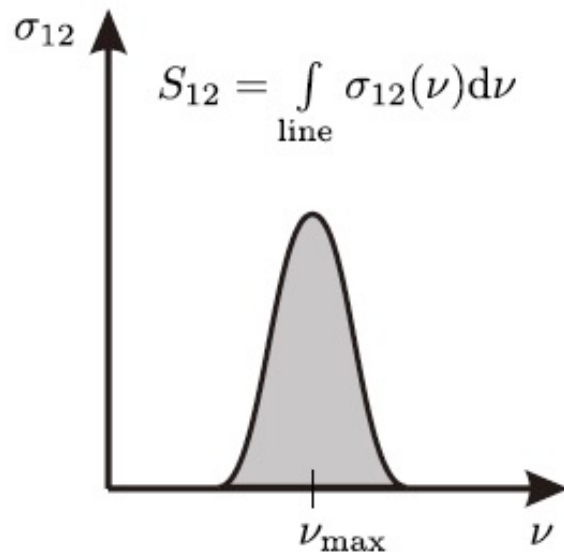
$$\rho(\nu) \approx \rho(\nu_{12}) \quad \longrightarrow \quad K_{12} \approx \frac{c}{h} \rho(\nu_{12}) \int_{\text{line}} \frac{\sigma_{12}(\nu)}{\nu} d\nu$$

Definition of integrated absorption cross section:

$$G_{12} = \int_{\text{line}} \frac{\sigma_{12}(\nu)}{\nu} d\nu$$

Definition of line strength:

$$S_{12} = \int_{\text{line}} \sigma_{12}(\nu) d\nu$$



These definitions lead to relations between measurable quantities and Einstein coefficients:

$$K_{12} \approx \frac{c}{h} \rho(\nu_{12}) G_{12}$$

and $B_{12} \approx \frac{c}{h} G_{12}.$

Relation between G and S:

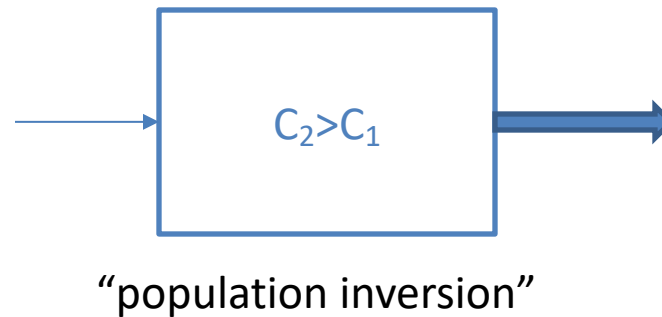
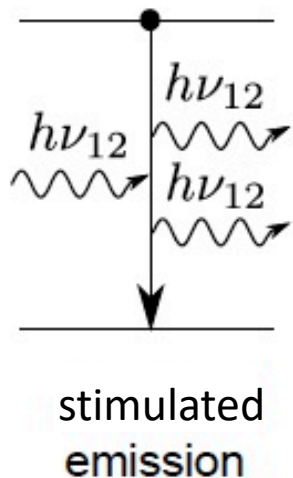
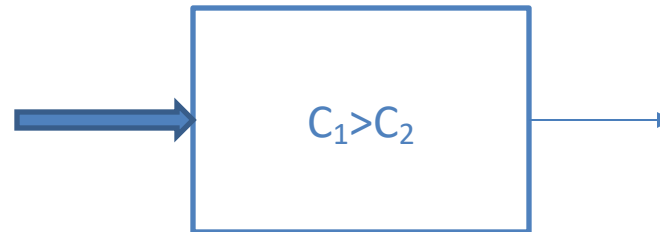
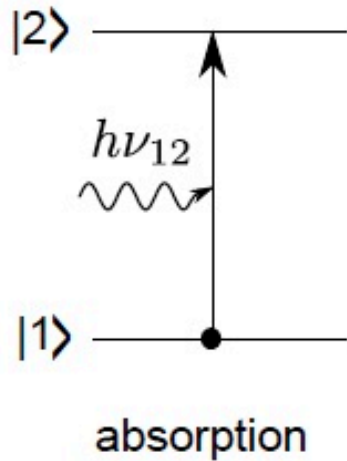
$$G_{12} \approx \frac{1}{\nu_{12}^{\max}} S_{12} \quad \longrightarrow \quad G_{12} \approx \frac{c^2}{8\pi\nu_{12}^3} A_{21}.$$

Reminder: Einstein coefficient can be expressed as function of transition moment

$$B_{12} = \frac{8\pi^3}{3h^2(4\pi\epsilon_0)} |M_{12}|^2 \quad M_{12} = \langle 2|\hat{\vec{\mu}}|1\rangle$$

 $G_{12} \approx \frac{8\pi^3}{3hc(4\pi\epsilon_0)} |M_{12}|^2$

1.5 Population inversion and laser operation



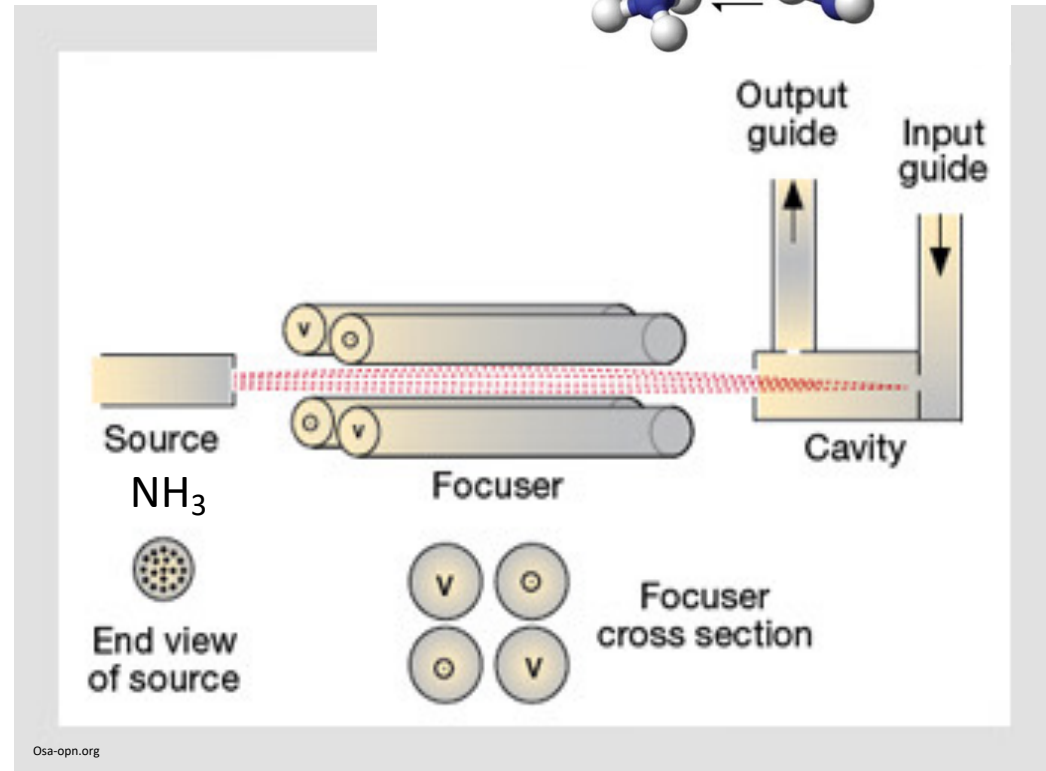
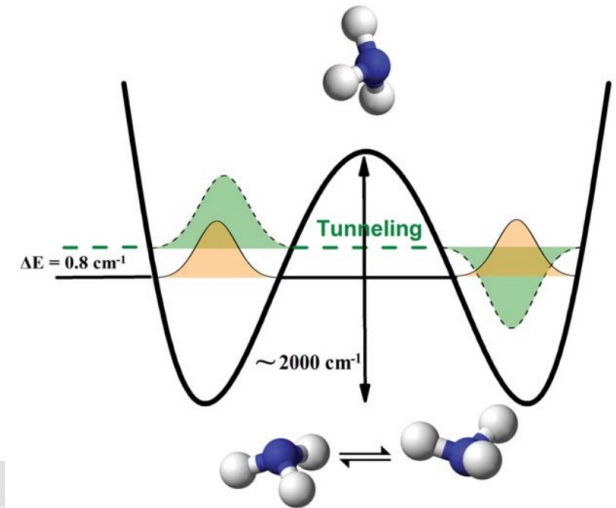
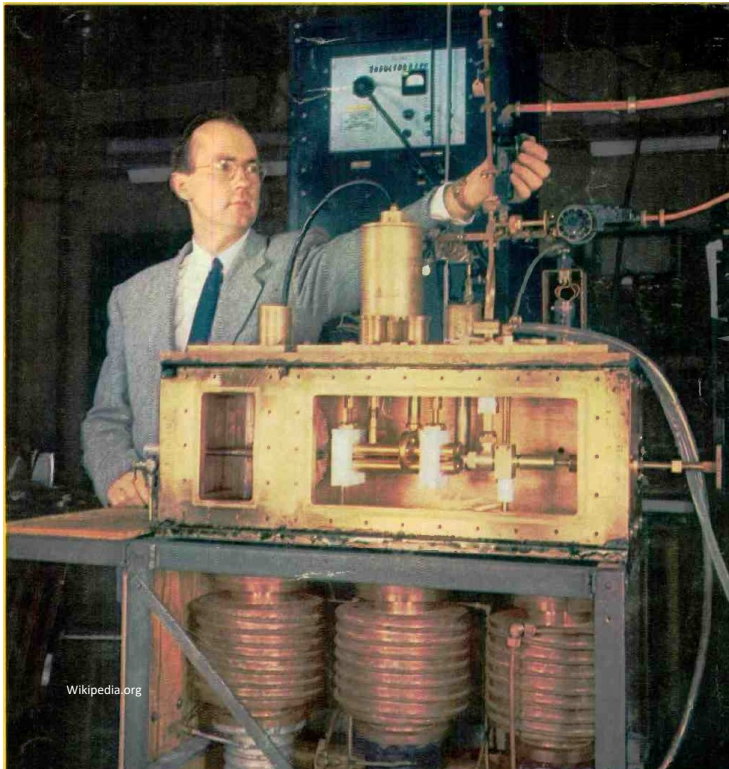
The net gain can be expressed as

$$\frac{I}{I_0} = \exp [\sigma_{12} (C_2 - C_1) \ell]$$

LASER: Light Amplification by Stimulated Emission of Radiation

First realization:

MASER, (M=microwave), Charles Townes, 1953



Principle of laser operation

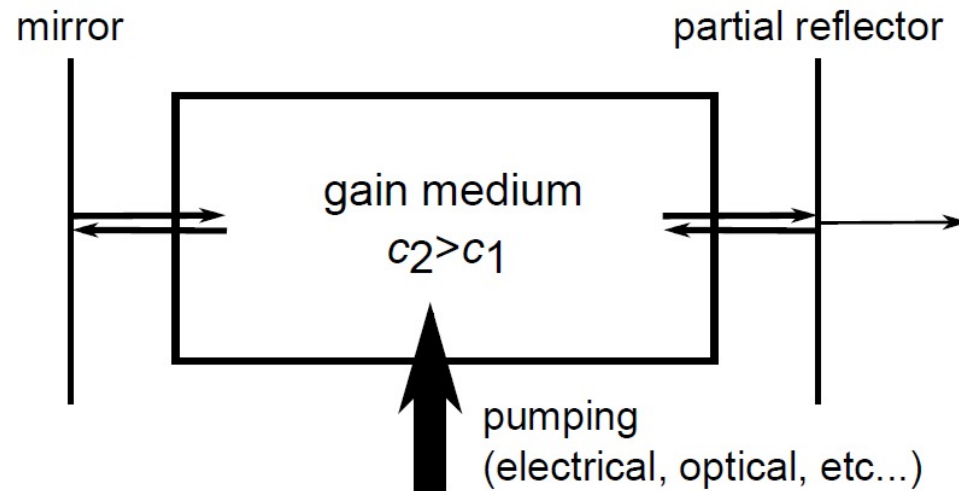


Figure 1.8: Experimental setup of a laser

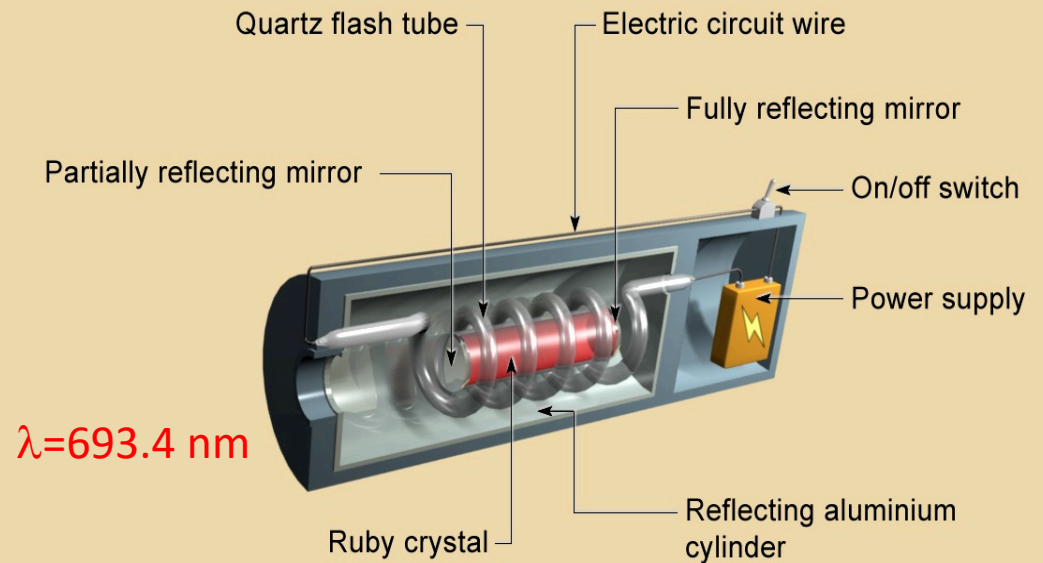
Light Amplification by Stimulated Emission of Radiation (LASER)

First realization:

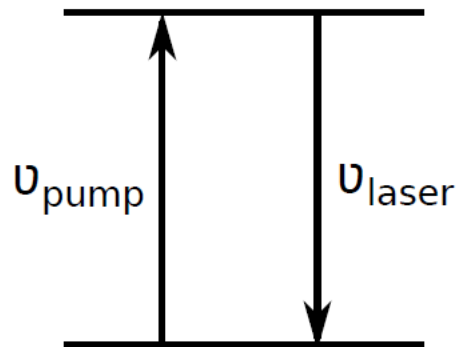
Theodore Maiman, 1960



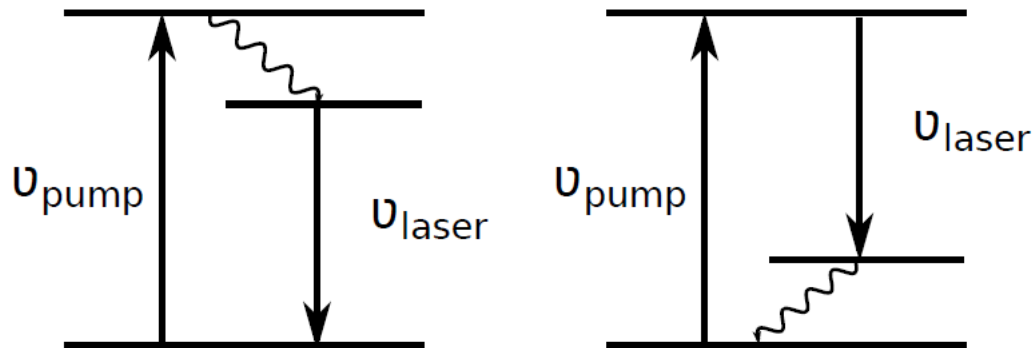
Cut-away View of a Ruby Laser



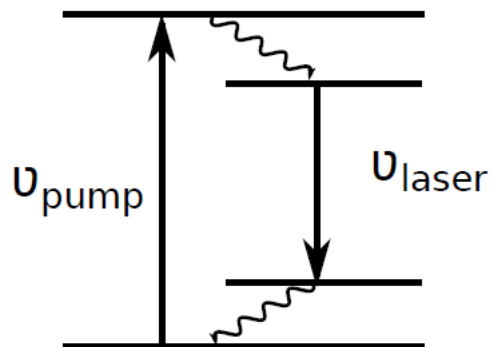
2-level laser



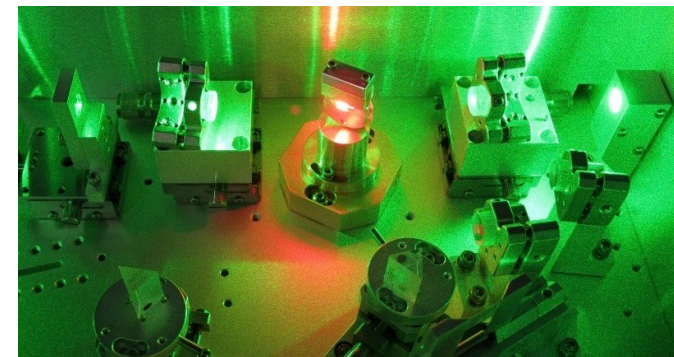
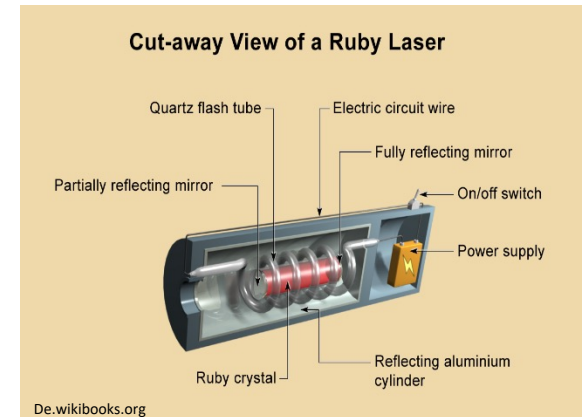
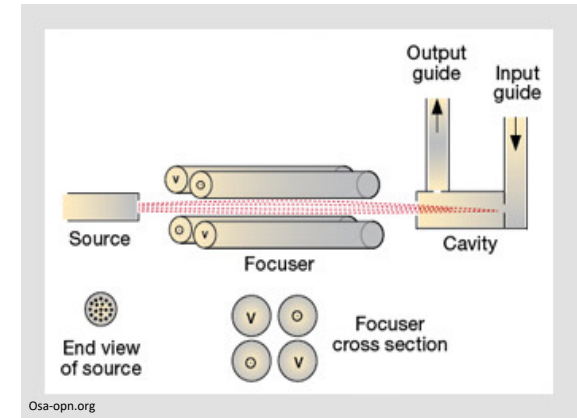
3-level laser



4-level laser



Examples



Ti:Sa laser (our laboratory)

Equations for laser operation

Rate of photon production:

$$\frac{dN_\gamma}{dt} = (N_2 - N_1) B_{21} \rho'(\nu_{12}) + A_{21} N_2$$

$$N_\gamma = \underbrace{g(\nu_{12})}_{\substack{\text{mode} \\ \text{density}}} \underbrace{\Delta\nu}_{\substack{\text{band-} \\ \text{width}}} \underbrace{n(\nu_{12})}_{\substack{\text{\# of photons} \\ \text{per mode}}}$$

$$\hookrightarrow \frac{dn(\nu_{12})}{dt} = (N_2 - N_1) \frac{B_{21} \rho'(\nu_{12})}{g(\nu_{12}) \Delta\nu} + \frac{A_{21}}{g(\nu_{12}) \Delta\nu} N_2$$

Since $V \rho'(\nu_{12}) = g(\nu_{12}) h \nu_{12} n(\nu_{12})$, using $g(\nu_{12}) = \frac{8\pi\nu_{12}^2}{c^3}$

$$V \rho'(\nu_{12}) = \frac{8\pi V}{c^3} \nu_{12}^2 h \nu_{12} n(\nu_{12})$$

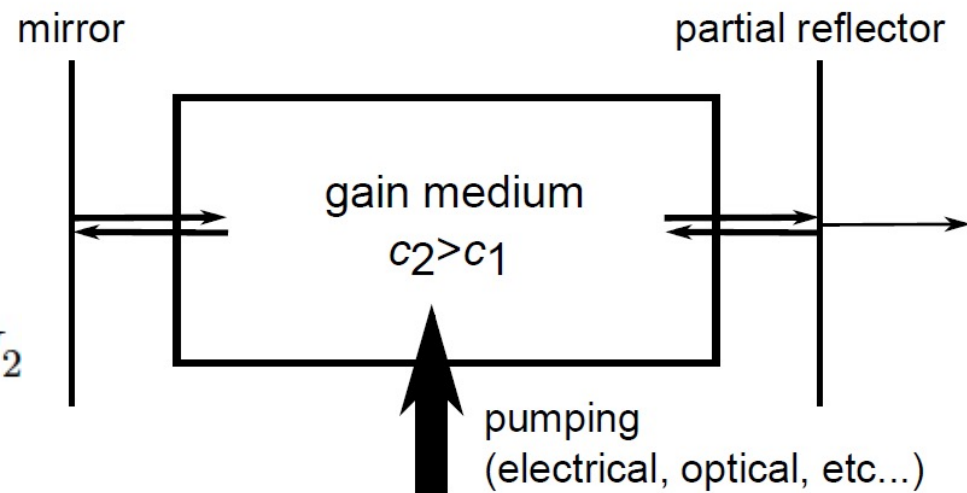


Figure 1.8: Experimental setup of a laser

and with the expressions for A_{21} and B_{21}

$$B_{21}\rho'(\nu_{12}) = \frac{A_{21}c^3}{8\pi h\nu_{12}^3}\rho'(\nu_{12}) = A_{21}n(\nu_{12})$$

With the definition

$$W = \frac{A_{21}}{g(\nu_{12})\Delta\nu}$$

we obtain the **laser equation**

$$\frac{dn(\nu_{12})}{dt} = (N_2 - N_1)Wn(\nu_{12},t) + WN_2$$

When the pump process is fast enough, one can assume that N_1 and N_2 reach a stationary regime

boundary condition $n(\nu_{12}) = 0$ at $t = 0$

$$n(\nu_{12},t) = \frac{N_2}{N_2 - N_1} (\exp(W(N_2 - N_1)t) - 1)$$

Losses from the cavity by, *e.g.* scattering or a partially transmitting mirror lead to an additional term in the laser equation

$$\frac{dn(\nu_{12})}{dt} = \underbrace{(N_2 - N_1) W n(\nu_{12}) + W N_2}_{\text{gain}} - \underbrace{k_1 n(\nu_{12})}_{\text{losses}}$$

For large values of $n(\nu_{12}, t)$ the contribution from spontaneous emission is often negligible, leading to the condition of laser operation

$$\frac{dn(\nu_{12})}{dt} > 0 \quad \text{or} \quad (N_2 - N_1) W > k_1, \quad (1.70)$$

Condition of operation for a laser: $\frac{N_2 - N_1}{V} \frac{c^3}{8\pi\nu^2} \frac{A_{21}}{\Delta\nu} > k_1$